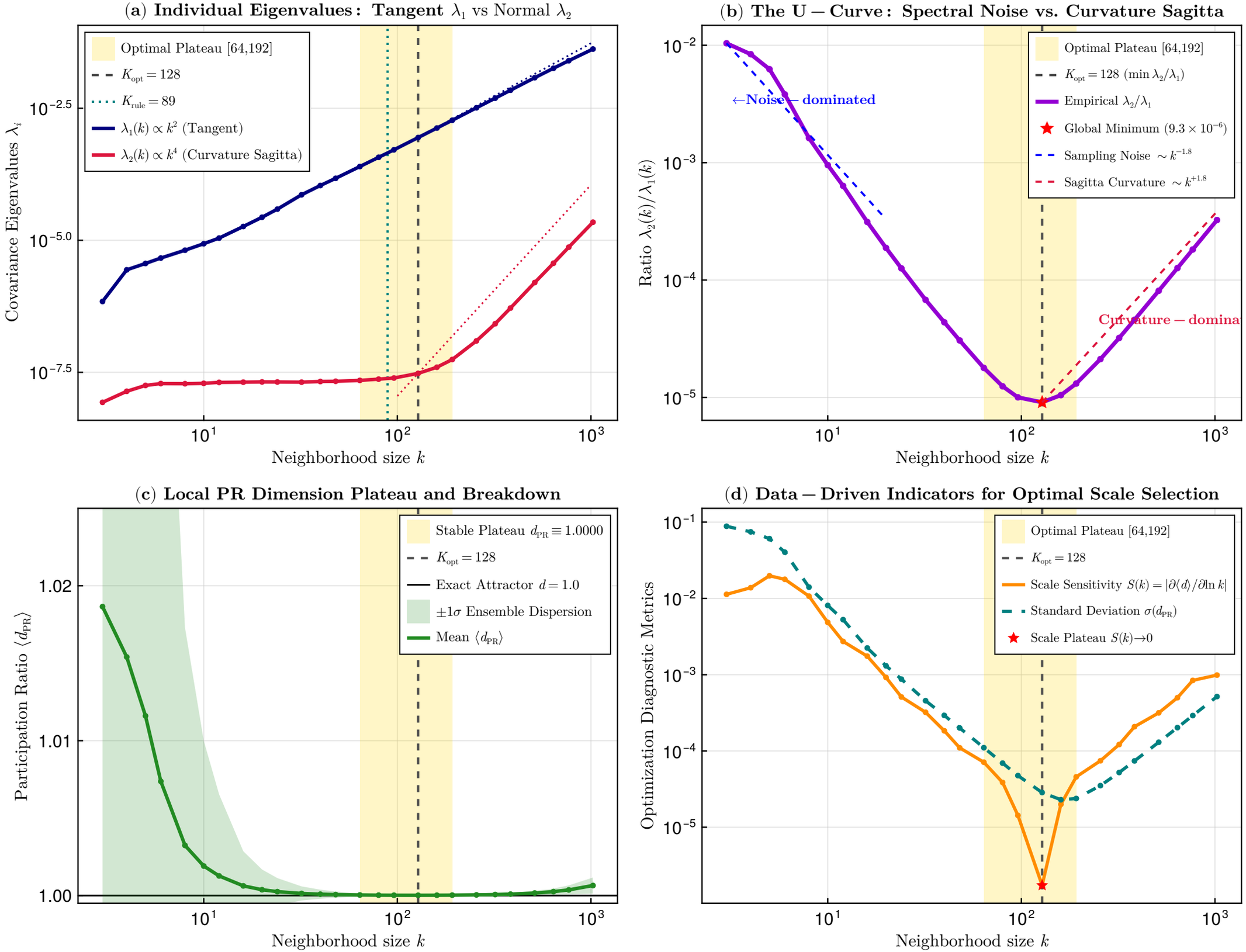

(a) Individual Eigenvalues: Tangent λ_1 vs Normal λ_2
Optimal Plateau [64,192]
$K_{opt} = 128$
$K_{rule} = 89$
$\lambda_1(k) \propto k^2$ (Tangent)
$\lambda_2(k) \propto k^4$ (Curvature Sagitta)
Covariance Eigenvalues λ_i
$10^{-2.5}$
$10^{-5.0}$
$10^{-7.5}$
10^1
10^2
10^3
Neighborhood size k
(b) The U — Curve: Spectral Noise vs. Curvature Sagitta
Optimal Plateau [64,192]
$K_{opt} = 128$ (min λ_2/λ_1)
Empirical λ_2/λ_1
Global Minimum (9.3×10^{-6})
Sampling Noise $\sim k^{-1.8}$
Sagitta Curvature $\sim k^{+1.8}$
←Noise — dominated
Curvature — domina
Ratio $\lambda_2(k)/\lambda_1(k)$
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^1
10^2
10^3
Neighborhood size k
(c) Local PR Dimension Plateau and Breakdown
Stable Plateau $d_{PR} \equiv 1.0000$
$K_{opt} = 128$
Exact Attractor $d = 1.0$
$\pm 1\sigma$ Ensemble Dispersion
Mean $\langle d_{PR} \rangle$
Participation Ratio $\langle d_{PR} \rangle$
1.02
1.01
1.00
10^1
10^2
10^3
Neighborhood size k
(d) Data — Driven Indicators for Optimal Scale Selection
Optimal Plateau [64,192]
$K_{opt} = 128$
Scale Sensitivity $S(k) = |\partial\langle d \rangle / \partial \ln k|$
Standard Deviation $\sigma(d_{PR})$
Scale Plateau $S(k) \to 0$
Optimization Diagnostic Metrics
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^1
10^2
10^3
Neighborhood size k